SOHAM KHAN

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CARRIER OBJECTIVE

Electronics and Communication Engineering undergraduate with a strong foundation in circuit design, RF engineering, and embedded systems. Aspiring to build a career in RF technologies integrated with VLSI and semiconductor systems, leveraging research, innovation, and engineering expertise to develop next-generation wireless and integrated circuit solutions.

EDUCATION

Institute of Engineering and Management, Salt Lake **Jul 2024 - Jun 2028**
Bachelor of Technology *Kolkata, India*
GPA: 7.88

Hem Sheela Model School **Apr 2019 - May 2023**
Senior Secondary, Science(PCMB) *Durgapur, India*

SKILLS SUMMARY

- **Programming & Development**: Python, C, APIs, Arduino IDE,Java
- **Domain Knowledge**:RF Engineering, Embedded Systems, PCB Design, Circuit Design, Electromagnetic Simulation, Semiconductor Devices & VLSI Fundamentals
- **Design & Simulation Tools**: Altium Designer, LTSpice, MATLAB, Simulink, CST Studio Suite,ANSYS HFSS
- **Platforms**: PyCharm, Jupyter Notebook, Visual Studio Code
- **Professional Skills**:Technical Documentation, Experimental Analysis,Technical Report Writing

WORK EXPERIENCE

Summer Research Intern ***May 2026 - July 2026***
IEEE MTT-S Region 10
- *Researching a 2.45 GHz ISM-band microwave radar system for non-contact respiratory monitoring.*
- *Designing and optimizing a rectangular microstrip patch antenna using CST Studio Suite for biomedical radar sensing.*
- *Investigating Doppler/FMCW radar techniques and DSP-based respiration signal processing for RF healthcare applications.*

Sanchar Mitra **Sep 2025 - Present**
Internship
- Telecom awareness & outreach, cybersecurity awareness, community engagement
- Developed skills in telecom fraud prevention, mobile security, and cybersecurity awareness.
- Gained experience in grassroots outreach and community engagement by working with local communities, NGOs, WB LSA and authorities; enhanced coordination and report-building skills for addressing telecom-related issues.

PROJECTS

AI-Based Surface AQI & HCHO Hotspot Identification using Satellite Data **May 2026 - Jul 2026**
ISRO Bharatiya Antariksh Hackathon 2026

Built an AI-powered framework for high-resolution AQI prediction and HCHO hotspot identification using multi-source satellite data
Developed a CNN–LSTM with Attention model integrating INSAT-3D, Sentinel-5P, CPCB, MODIS, and ERA5 datasets for spatiotemporal environmental analysis.
Tech Stack: Python, TensorFlow, Google Earth Engine, Streamlit, GitHub

AI Powered Soil Health Monitoring System-Google Solution Challenge 2025 **Dec 2024 - Feb 2025**
- Achieved a 96% accuracy rate in forecasting student academic performance by developing and deploying a machine learning model.
- Managed data integrity by handling missing values and encoding categorical variables, enhancing data quality by 33%.
- Conducted experiments with both classification and regression algorithms to identify the most suitable approach

Bi-Directional Gate Counter using Arduino UNO **Jul 2024 - Sep 2024**
- Developed a bi-directional visitor counter system using Arduino Uno and IR sensors to track people entering and exiting a space.
- Designed and programmed the system in Embedded C/Arduino IDE, optimizing sensor accuracy and response time.
- Demonstrated applications in classrooms, offices, and public halls for efficient resource utilization, crowd management, and energy savings (e.g., automating lights/fans based on occupancy).

CERTIFICATES

- Fundamentals of Semiconductor Devices,NPTEL,IISc Bangalore — April 2026
- Internet of Things and AI Cloud Specialization - University of California San Diego — October 2025
- McKinsey.org Forward Program: Issued by McKinsey & Company — July 2025
- Postman API Fundamentals Student Expert by Postman — March 2025
- Programming with Generative AI - IIT Guwahati — December 2024
- MATLAB Onramp by MathWorks — October 2024